Joan Alcaide Núñez

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Research Interests

Theoretical Cosmology, Dynamical Dark Energy, Fundamental Physics

Education

B.Sc. Physics 2025 – present
Ludwig-Maximilians-Universität München (LMU Munich) *München, Germany*
- Status: Semester 3

Dual German-Spanish High School 2017 – 2025
German School of Barcelona *Barcelona, Spain*
- Final Abitur GPA: 1.0/1.0
- Thesis: Recomputations of the Hubble-Lemaître Constant with Distance Ladder Methods in Consideration of the Current Crisis in Cosmology PDF (in German)

Research Experience

Research Stay - GRB Cosmology Summer 2025
Osservatorio Astronomico di Brera (OAB-INAF) *Brera, Italy*

- Cosmological parameter inference $(\Omega_m, \Omega_\Lambda, H_0)$ with χ^2 statistics of GRB Amati relation

Founder and Member of GAMMA Team 2024 – 2026
CAPIBARA Collaboration *International*

- Developing γ-ray transient monitoring CubeSat mission architecture and proposal, presented at 5th SSEA
- Founding and administrating the collaboration of high school students learning about the high-energy multi-messenger universe

2-week Internship - Earth Observation Autumn 2024
Earth Observation Center (EOC-DLR) *Munich, Germany*

- Analyzed thermal images of LandSat-8 with GDAL
- Intership awarded in the Jugend Forscht competition

Research Stay - Infrared Ia Supernova Photometry Summer 2024
Institute of Space Sciences (ICE-CSIC-IEEC) *Barcelona, Spain*

- Measuring distances with SNe photometry using SOFI (ESO) data and the ZTF and ATLAS light curves
- Scientific programming in Python (numpy, pandas scipy, astropy, ...) and LaTeX

Research Project - Cepheid Cosmology Winter 2023

- Recomputed the Hubble-Lemaître Constant using modern Cepheid and redshift data $(H_0 = 70.26 \pm 7.1 \mathrm{kms}^{-1}\mathrm{Mpc}^{-1})$
- This project was part of the first phase of the Youth and Science fellowship

Awards

Youth & Science Fellowship *2023 – 2025*
3-year funded fellowship for research stays internationally for selected high school students
2025 Distinction for extraordinary performance in high school physics

IAAC *2025*
Silver Honour and National Award (Spain) (top 7% of $>$ 12000 students worldwide), similar results in 2023 and 2024

Jugend Forscht *2025*
First Prize Iberia Competition (2024 & 2025), First prizes in regional and national rounds, Special Award in Scientific Photography

Outreach & Teaching

Poster Talk *Munich, Germany, 2026*
Symposium on Space Educational Activities

Talk CAPIBARA Collaboration *Barcelona, Spain, 2025*
CosmoXarxa Event at Barcelona's Science Museum

2-session Introduction to Astrophysics *Barcelona, Spain, 2025*
11th graders at German School of Barcelona

Private Tutor *Barcelona, Spain, 2024-2025*
Mathematics, Physics, German, English

Outreach Activity on Space Exploration *Spain, 2023*
4th graders at public school

Explainer Volunteer *Barcelona, Spain, 2022-2023*
Barcelona's Science Museum CosmoCaixa

Technical Skills

- **Programming**: Python, Julia, C++, Shell Scripting (basic).
- **Frameworks**: Git, LaTeX, Markdown, HTML/CSS, Linux, NeoVim, VSCode.
- **Languages**: English (fluent), German (fluent), Spanish (native), Catalan (native).

Work Experience

- **HiWi Job (Statistics, forms, data analysis)**: LMU Munich Research Funding Unit & Central Diversity Management